

Open Cooperative Perception System

(O-CPS)

Technical Specification and Whitepaper

Strynex Reference Architecture

Draft v0.2

April 2026

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No patent claims are asserted against good-faith implementers.

Abstract

The Open Cooperative Perception System (O-CPS) is an open, manufacturer-agnostic cooperative perception and control layer for road vehicles and roadside infrastructure. It reuses existing V2X radio, network, and security standards (C-V2X PC5 / NR-V2X sidelink, ITS-G5, SAE J2735, SAE J3161, ETSI TS 103 324, IEEE 1609.2, ETSI TS 103 097) and adds three primitives that the current standards ecosystem leaves underspecified: a normalized perception profile that allows vehicles with radically different sensor suites to exchange a common world model; a fusion and trust framework that produces calibrated confidence in multi-source object tracks; and a Cooperative Adaptive Cruise Control (C-ACC) Group Protocol that allows driver-authorized platoons to form, persist, hand off leadership, and dissolve without cloud dependence.

This document is the v0.2 specification draft. It is the successor to the v0.1 draft circulated in August 2025. Relative to v0.1 it adds: a complete wire-format schema for the full message set; a formal trust model with continuous attestation and misbehavior revocation; a safety architecture aligned to ISO 26262 with explicit ASIL allocation per message class; a conformance-class framework; and the Slipstream Mode overlay introduced as an optional application-layer extension.

O-CPS is a standards and influence play, not a product. It is published in the open, with a permissive license posture, to give automakers, suppliers, municipalities, and fleet operators a shared target to converge on rather than a vendor-specific lock-in.

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1. Introduction

Modern vehicles are increasingly capable of perceiving their immediate environment with high fidelity. Radar, lidar, camera arrays, and ultrasonic sensors feed local perception stacks that maintain a rich model of the world inside a single vehicle. The limit of that model is the vehicle's own sensor envelope: roughly 200 meters of useful range, heavily degraded by weather, occlusion, and geometry.

Cooperative perception breaks that limit. When vehicles share their local perception with each other and with roadside infrastructure, each participant gains a world model that extends far beyond its own sensors, sees around corners, and persists through weather and occlusion. The safety and efficiency gains are well-understood in the literature and are the subject of active standards work at SAE, ETSI, IEEE, and 3GPP.

The problem is not the absence of standards. The problem is that the existing standards ecosystem is fragmented across radio layers, message sets, security frameworks, and regional profiles, and leaves several practical questions unanswered. How do vehicles with different sensor suites produce interoperable perception output? How is trust established and revoked in real time without a continuously-available backhaul? How do driver-authorized cooperative behaviors (platoons, merges, intersection assist) form and dissolve safely?

O-CPS answers these questions by composing existing standards where they are adequate and specifying the missing pieces in a single coherent profile. It is deliberately thin. It adds the minimum necessary primitives on top of the existing V2X stack and refuses to invent new radio layers, new cryptographic primitives, or new market structures.

1.1 Design Heritage

O-CPS draws directly on three bodies of prior work. From SAE J2735 and J3161 it inherits the BSM/PSM message family and the PC5-based safety messaging conventions. From ETSI TS 103 324 it inherits the Collective Perception Message (CPM) concept and the perceived-object container model. From IEEE 1609.2 and ETSI TS 103 097 it inherits the pseudonym certificate and misbehavior-reporting frameworks. None of these are wholly sufficient in isolation; O-CPS is the profile that binds them together and fills the gaps.

1.2 Relationship to Strynex

“Strynex” is the reference implementation and vehicle-facing brand for an O-CPS stack. The two names are deliberately distinct: O-CPS is the open specification that any automaker, supplier, or municipality may implement without royalty; Strynex is one concrete implementation and a set of optional application-layer overlays (notably Slipstream Mode, §13) that sit on top of the base profile. A vehicle may advertise O-CPS conformance without implementing any Strynex-branded overlay, and the protocol MUST continue to interoperate correctly.

2. Terminology and Conventions

The key words MUST, MUST NOT, REQUIRED, SHALL, SHALL NOT, SHOULD, SHOULD NOT, RECOMMENDED, MAY, and OPTIONAL in this document are to be interpreted as described in RFC 2119 when, and only when, they appear in all capitals.

The following terms are used throughout:

Term	Definition
Participant	Any entity that transmits or consumes O-CPS messages. Includes vehicles, roadside units (RSUs), and mobile clients (e.g., phones acting as vulnerable-road-user beacons).
Vehicle	A participant with longitudinal and/or lateral control authority.
RSU	Roadside Unit. A fixed participant, typically maintained by a road authority, that transmits infrastructure messages (SPaT, MAP, environmental) and may relay or fuse nearby vehicle messages.
VRU	Vulnerable Road User. A non-vehicle participant (pedestrian, cyclist) that may transmit presence and may consume relevant messages, but has no control authority.
Perception Object	A single entity detected by a participant's sensor fusion: class, pose, kinematic state, confidence, and provenance.
World Model	A participant's current estimate of the set of Perception Objects in its area of interest, fused from its own sensors and received O-CPS messages.
Pseudonym Certificate	A short-lived identity certificate issued by an SCMS or equivalent CA, used to authenticate messages without linking them to a persistent vehicle identity.
C-ACC Group	A set of two or more vehicles operating under the Cooperative Adaptive Cruise Control Group Protocol (§13.1).
Leader	The vehicle in a C-ACC Group whose trajectory is used as the coordination anchor.
Attestation	A signed claim about the integrity of a participant's software and sensor stack, used by peers to calibrate trust in that participant's messages.

3. Scope and Goals

3.1 In Scope

- Message formats and wire encoding for cooperative perception, intent, group membership, and attestation.
- A normalized perception profile that allows heterogeneous sensor suites to produce interoperable output.
- A trust and security model covering identity, pseudonymity, continuous attestation, misbehavior detection, and revocation.
- A privacy model covering what data may be transmitted, under what conditions, and for how long it may be retained by any participant.
- A fusion framework that produces calibrated confidence in multi-source object tracks.
- Cooperative behaviors: the C-ACC Group Protocol and the optional Slipstream Mode overlay.
- Safety architecture and ISO 26262 ASIL allocation per message class.
- Vehicle integration interfaces to ADAS, gateway ECUs, and telematics units.
- Conformance classes and a self-test framework.

3.2 Non-Goals

O-CPS explicitly does not specify:

- New radio physical layers. O-CPS runs on C-V2X PC5 sidelink (LTE-V2X Rel-14/16 and NR-V2X Rel-16+) and on ITS-G5 (IEEE 802.11p / bd) where available. No new PHY is defined.
- New cryptographic primitives. Signatures use ECDSA over P-256 (MUST) and may additionally use Ed25519 (MAY) as defined in IEEE 1609.2-2022 and ETSI TS 103 097 v2.1.1.
- Cloud-dependent control paths. O-CPS is a sidelink-first protocol. Cloud interfaces are OPTIONAL and MUST NOT be in any safety-critical loop.
- In-vehicle HMI design. How a participant surfaces cooperative state to the driver is out of scope. How it gates driver authorization is in scope (§13.1.3).
- Commercial licensing of sensor hardware, compute platforms, or backend services.

3.3 Goals

An O-CPS implementation, conformant to this specification, is expected to achieve the following properties. These are design goals, not requirements; the normative requirements that realize them appear in later sections.

- Interoperability across vendors with no prior coordination, provided both sides are conformant to the same conformance class.
- Bounded latency: a PerceptionSummary observed at one participant is available at peer participants within 100 ms at the 99th percentile under nominal channel conditions.
- Graceful degradation: loss of cooperative messages MUST result in a return to local-sensor-only behavior, not in unsafe actuation.

- Cryptographic deniability of individual vehicles' long-term identity: no O-CPS message, or correlated set of messages, shall reliably identify a specific vehicle over time without possession of a revocation key held by the issuing CA.
- Retrofittability: the baseline conformance class (Class A, §17.1) **MUST** be implementable on current C-V2X/ITS-G5 hardware without ECU replacement.

4. Design Philosophy

O-CPS is opinionated. The opinions below are load-bearing: a profile that relaxed any of them would be a different profile, not a variant of this one.

4.1 Driver Authority is Absolute

No cooperative behavior specified in this document takes control authority away from the driver. Every automated longitudinal or lateral response derived from O-CPS messages **MUST** be implemented as an advisory or as a conditional input to an ADAS function that the driver can override instantaneously. Any deviation is a conformance failure.

This is not a philosophical preference. It is the only posture under which regulators in every major jurisdiction will permit cooperative perception to influence vehicle behavior in the near term, and it is the posture that limits the blast radius of any compromise or misbehavior.

4.2 Sidelink First

Safety-relevant O-CPS messages travel vehicle-to-vehicle or vehicle-to-infrastructure over PC5 sidelink or ITS-G5 direct, never through a cellular core or cloud service. Cloud-mediated paths exist in this specification (e.g., credential provisioning, misbehavior report aggregation) but are deliberately confined to non-safety-critical functions. A vehicle that loses cellular connectivity **MUST** continue to operate all safety-critical cooperative behaviors without degradation.

4.3 Minimize, Localize, Expire

O-CPS transmits the minimum data necessary for a coherent shared world model. No personally identifiable information is placed on the wire. Data is localized: a PerceptionSummary is addressed to geographic scope, not to a population of subscribers. Data expires: every message has a bounded validity window, and receivers **MUST** discard data older than its stated lifetime.

4.4 Trust is Continuous

A valid certificate at session start is necessary but not sufficient. Participants continuously attest to the integrity of their software and sensor stack (§10.2), and peers calibrate the weight they give to a participant's messages based on attestation freshness, consistency with local sensors, and history. A vehicle whose attestation has lapsed or whose reports repeatedly diverge from consensus **MUST** be downgraded to advisory-only weight, and eventually reported to the misbehavior authority.

4.5 Fail Closed, Degrade Gracefully

Every cooperative behavior specified in this document has an explicitly defined failure state that reverts to local-sensor-only operation. There is no failure mode that requires continued cooperation to remain safe. A participant that cannot maintain O-CPS conformance (due to software fault, sensor fault, certificate expiry, or misbehavior) **MUST** cease transmission and **MUST NOT** consume peer messages for control inputs until conformance is restored.

4.6 No Market Structure

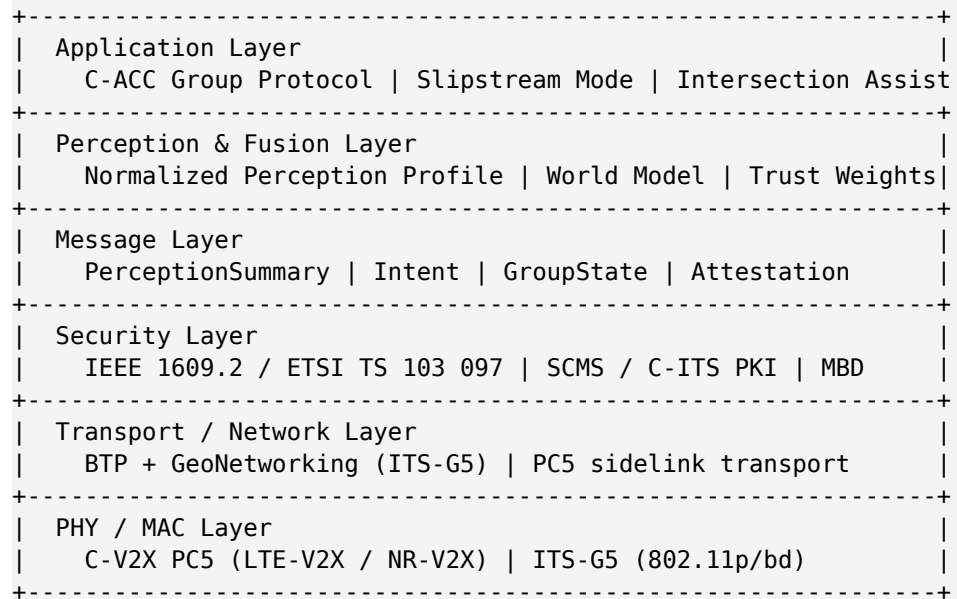
O-CPS defines no platform. There is no central operator, no mandatory service tier, no per-message fee, and no subscription gate for any function specified in this document. Automakers

and operators MAY build commercial services on top of O-CPS; they MUST NOT condition participation in the base profile on such services.

5. System Architecture Overview

An O-CPS participant is organized as a layered stack. The stack is conceptually identical in vehicles and in RSUs; implementations differ in which layers are populated and which interfaces are exposed to external control.

5.1 Stack Diagram



5.2 Layer Responsibilities

Layer	Responsibilities	Normative References
PHY / MAC	Radio transmission and reception; channel access; congestion control.	3GPP TS 36.213, TS 38.213; IEEE 802.11p/bd
Transport / Network	Geocast routing (ITS-G5), sidelink transport (PC5), message segmentation and reassembly.	ETSI EN 302 636 series; 3GPP TS 23.285
Security	Message authentication, pseudonymity, misbehavior detection, revocation.	IEEE 1609.2-2022; ETSI TS 103 097 v2.1.1
Message	O-CPS wire formats and encoding (§9).	This document
Perception & Fusion	Normalized perception profile, world-model maintenance, trust-weighted fusion.	This document; ETSI TS 103 324 (informative)
Application	Cooperative behaviors that consume the world model and produce advisory or control outputs.	This document

5.3 Interfaces

An O-CPS vehicle implementation exposes the following internal interfaces:

- **V-PERCEPT:** inbound interface from the vehicle's local perception stack. Consumes object lists and pose estimates in the implementation-specific format of the onboard ADAS.
- **V-CTRL:** outbound advisory interface to the vehicle's ADAS/control stack. Produces gap targets, lane-keeping biases, and Slipstream state transitions. Never authoritative.
- **V-HMI:** outbound interface to the driver-facing HMI. Surfaces cooperative state and solicits driver authorization where required.
- **V-CRED:** interface to the vehicle's hardware security module (HSM) for private-key operations and attestation. MUST be protected by secure boot and SecOC.
- **V-DIAG:** diagnostic and logging interface. Emits signed diagnostic records for fleet-owner retrieval; does not expose any externally-transmitted message contents.

6. Layered Stack Detail

6.1 PHY / MAC

O-CPS is radio-agnostic within the constraints of regional regulation. Implementations **MUST** support at least one of: C-V2X PC5 (either LTE-V2X as specified in 3GPP Rel-14/16 or NR-V2X as specified in Rel-16 and later), or ITS-G5 as specified in IEEE 802.11p or its successor 802.11bd. Implementations intended for deployment in North America **SHOULD** prioritize C-V2X PC5 in Channel 183 (5.895–5.925 GHz) consistent with the FCC’s 2020 allocation. Implementations intended for deployment in the European Union **SHOULD** support both C-V2X and ITS-G5 on channels in the 5.875–5.905 GHz range per ETSI EN 302 571.

Dual-stack implementations **MUST** listen on all supported radios simultaneously and **MUST NOT** transmit duplicate application-layer messages across both radios without explicit congestion-control coordination; the decongestion algorithm specified in ETSI TS 103 175 applies.

6.2 Transport / Network

Over ITS-G5, O-CPS messages are carried in GeoNetworking packets with BTP-B (non-interactive) headers, per ETSI EN 302 636-4 and EN 302 636-5. Messages are addressed to a geographic destination area, typically a disc or rectangle centered on the transmitter and sized to the message’s relevance horizon (§9.1.4).

Over C-V2X PC5, messages are carried in sidelink bearers with ProSe Per-Packet Priority (PPPP) values assigned per message class (Table 9-2). NR-V2X implementations **MAY** use HARQ feedback as specified in 3GPP TS 38.321 for PerceptionSummary and Attestation messages; **MUST NOT** use HARQ for Intent messages, which have sub-100 ms deadlines that do not tolerate retransmission.

6.3 Security

Every O-CPS message that is transmitted over the air **MUST** be signed. The signature covers the entire message payload, including the timestamp field. Signatures are ECDSA over the NIST P-256 curve, computed over the SHA-256 hash of the payload. Signing keys are held in an HSM conformant to at least FIPS 140-3 Level 2 or Common Criteria EAL 4+.

Signing certificates are pseudonym certificates issued by a Security Credential Management System (SCMS) per IEEE 1609.2.1 or by a C-ITS PKI per ETSI TS 102 940 and TS 102 941. Pseudonym certificates have a maximum validity of one week and are rotated at a minimum interval specified in §10.1.2. Rotation uses butterfly key expansion per IEEE 1609.2.1 to provide unlinkability across certificate epochs.

6.4 Message

The message layer is specified in full in §9. It defines five message classes: PerceptionSummary, Intent, GroupState, Attestation, and Telemetry. All messages share a common header containing a timestamp, a pseudonym-certificate identifier, a message class identifier, and a sequence counter.

6.5 Perception and Fusion

The perception and fusion layer consumes incoming PerceptionSummary and Intent messages and fuses them with the vehicle's local sensor output to produce a world model. Fusion is trust-weighted: the contribution of each source to a fused track depends on (a) the attestation state of the source, (b) the historical agreement between the source and local sensors, and (c) the self-reported confidence in the source message. The exact weighting function is implementation-defined, subject to the constraints in §12.

6.6 Application

The application layer hosts cooperative behaviors. This specification defines the C-ACC Group Protocol (§13.1) as a normative behavior. It defines Slipstream Mode (§13.2) and Intersection Assist (§13.3) as optional extensions. Other behaviors MAY be defined by extension profiles and MUST NOT conflict with the normative requirements of this document.

7. Normalized Perception Profile

Vehicles today ship with wildly different sensor suites. A current-generation luxury sedan may carry four radars, a lidar, six cameras, and twelve ultrasonics; a compact EV may carry one forward radar and three cameras; a commercial truck may carry a different superset again. Each suite produces a different native representation of the world: different coordinate frames, different object-class taxonomies, different confidence semantics, different update rates.

The Normalized Perception Profile is the contract that lets all of these produce interoperable output. It is not a least-common-denominator. It is a structured downscale that preserves the information a cooperative peer actually needs and discards the rest.

7.1 Reference Frame

All pose and kinematic data in O-CPS messages **MUST** be expressed in the World Geodetic System 1984 (WGS-84) frame, with latitude and longitude in decimal degrees, altitude in meters above the WGS-84 ellipsoid. Heading (yaw) **MUST** be expressed in degrees clockwise from true north. Velocity **MUST** be expressed in meters per second along the heading axis. Deviation from WGS-84 is a conformance failure.

7.2 Object Taxonomy

Objects in PerceptionSummary messages **MUST** be classified into exactly one of the following normalized classes. An implementation **MAY** maintain a richer internal taxonomy and collapse to this set on the wire.

Class ID	Name	Description
0x00	UNKNOWN	An obstacle of indeterminate type. MUST be used when the source cannot confidently classify.
0x01	PASSENGER_CAR	Light passenger vehicle, including SUVs and light pickups.
0x02	COMMERCIAL_TRUCK	Medium- or heavy-duty commercial vehicle.
0x03	BUS	Passenger bus (transit or motorcoach).
0x04	MOTORCYCLE	Two-wheeled motor vehicle.
0x05	BICYCLE	Non-motorized or e-bike bicycle with rider.
0x06	PEDESTRIAN	Human outside any vehicle.
0x07	ANIMAL	Large animal in or near the roadway.
0x08	STATIC_OBSACLE	Non-vehicle, non-living obstacle (debris, cone, barrier).
0x09	EMERGENCY_VEHICLE	Vehicle with active emergency signaling. MUST be asserted only on direct evidence (e.g., light-bar detection) or on receipt of a corresponding J2735 EVA message.
0x0A	CONSTRUCTION_E	Non-roadgoing equipment present in or near the

Class ID	Name	Description
	QUIPMENT	roadway.
0x0B-0xFE	RESERVED	Reserved for future assignment by the O-CPS registry.
0xFF	VENDOR_EXTENSION	Vendor-specific class. MUST be accompanied by a vendor OID. Peers MUST ignore if OID is unrecognized.

7.3 Confidence Semantics

Every object in a PerceptionSummary carries a confidence score in the range [0, 255] with the following semantics:

Range	Label	Interpretation
0–63	LOW	Source has weak evidence (single modality, short dwell, high ambiguity). Peers MUST NOT use as sole input for safety-critical advisory.
64–127	MEDIUM	Single-modality or short-dwell detection with corroboration. May be used as supporting evidence for fusion.
128–191	HIGH	Multi-modality fusion or extended dwell. Suitable as primary input to cooperative behaviors.
192–255	VERY_HIGH	Multi-modality fusion with independent corroboration and stable track. The source MUST have demonstrated track stability for at least 500 ms.

Confidence semantics are uniform across implementations: a source reporting HIGH confidence on a P-256 / bicycle-class track at range 40 m MUST have subjected that detection to the same evidentiary standard regardless of which sensor modalities it used. Systematic miscalibration (e.g., a vendor routinely reporting VERY_HIGH on single-modality camera-only detections) is detectable through consensus comparison (§10.3) and is a basis for misbehavior reporting.

7.4 Relevance Horizon

Every PerceptionSummary is scoped to a relevance horizon, expressed in meters. Peers MUST NOT consume objects beyond the declared horizon, and transmitters MUST NOT declare a horizon larger than the effective range at which their sensor suite can produce detections at MEDIUM confidence or better. Typical values:

- Urban intersection: 50–100 m
- Arterial road: 100–200 m
- Limited-access highway: 200–400 m
- RSU-sourced perception: up to 500 m, provided the RSU can substantiate the horizon in its conformance report.

8. Message Layer: Message Types and Wire Format

O-CPS defines five message classes. All messages share a common header and are encoded with Canonical Protocol Buffers, schema version 3, using the deterministic serialization rules of protobuf-3.21 or later. The use of protobuf is not a commitment to Google's implementation; any conformant encoder/decoder is acceptable.

The complete schemas appear in Appendix A. This section describes the classes in narrative form and specifies their rate and priority profiles.

8.1 Common Header

Every O-CPS message begins with an O-CPS Header:

```
message OcpsHeader {
  uint32 version          = 1; // 0-CPS schema version; MUST be 2 for this spec
  uint64 timestamp_ms     = 2; // ms since Unix epoch, UTC
  bytes  cert_id          = 3; // 8-byte identifier of the signing pseudonym cert
  uint32 msg_class        = 4; // 0x01=PerceptionSummary, 0x02=Intent,
                               // 0x03=GroupState, 0x04=Attestation,
                               // 0x05=Telemetry
  uint32 seq              = 5; // monotonically increasing per-sender counter
  uint32 horizon_m        = 6; // relevance horizon (meters)
  uint32 lifetime_ms      = 7; // message validity lifetime after timestamp
}
```

8.2 Rate and Priority Profile

The following table specifies the per-class rate ceilings, size envelopes, priority assignments, and ASIL allocations. These are normative.

Class	Rate	Size	PC5 PPPP	ITS-G5 AC	ASIL
PerceptionSummary	10–20 Hz	200–500 B	3	AC_VO	ASIL B
Intent	2–10 Hz	60–120 B	2	AC_VO	ASIL C
GroupState	1–5 Hz	120–240 B	3	AC_VI	ASIL C
Attestation	0.1–0.5 Hz	180–320 B	5	AC_BE	QM
Telemetry	≤ 0.1 Hz	≤ 1 KB	7	AC_BE	QM

Notes on the allocation: PerceptionSummary is ASIL B because a corrupted perception message could induce an inappropriate advisory gap target, but the integrator's ADAS retains ASIL D authority for the actual actuation, which limits upstream allocation. Intent is ASIL C because an Intent message can trigger an actuation-adjacent behavior (a driver-authorized lane change advisory) whose misstatement has higher consequence. GroupState is ASIL C because leader-election faults inside a platoon can induce longitudinal oscillation. Attestation and Telemetry are QM (Quality Management) because their loss degrades trust calibration and diagnostics but does not, in isolation, produce an unsafe actuation.

8.3 PerceptionSummary

PerceptionSummary is the workhorse message of O-CPS. Each message conveys the transmitter's own pose and a list of the objects it currently perceives. Typical transmission rate is 10 Hz in free flow, rising to 20 Hz during maneuvers or in dense traffic, and falling to 2 Hz when stationary in a non-roadway context (e.g., parked).

The message contains:

- The OcpsHeader.
- The transmitter's own pose (lat, lon, alt, yaw, pitch, roll, speed_mps, longitudinal acceleration).
- The transmitter's dimensions (length, width, height) in centimeters.
- A list of PerceivedObjects (0–N, N bounded by message-size envelope). Each PerceivedObject contains: class ID, confidence, relative position (range and bearing from transmitter), bounding box (length, width, height), velocity relative to transmitter, and a per-object source sensor mask (bitfield: radar, lidar, camera, ultrasonic, other).
- The ECDSA-P256 signature over the preceding fields.

8.4 Intent

Intent is the short-horizon prediction of what the transmitter is about to do. It is used by peers to anticipate maneuvers before they are executed and to coordinate during cooperative behaviors. Transmission is event-driven (on maneuver planning) with a background heartbeat rate of 2 Hz.

The message contains:

- The OcpsHeader.
- An intent code: one of STRAIGHT, LANE_CHANGE_LEFT, LANE_CHANGE_RIGHT, TURN_LEFT, TURN_RIGHT, MERGE, EXIT, STOP, YIELD, PLATOON_FORM, PLATOON_LEAVE.
- A prediction window (ms) over which the intent is valid.
- A confidence score (driver-committed vs. system-predicted).
- An optional GroupState reference (if the intent is group-coupled).
- The signature.

8.5 GroupState

GroupState is exchanged among C-ACC Group members to maintain a shared view of membership, ordering, leader identity, and operational mode. Transmission is 1–5 Hz, event-driven on state changes.

The message contains:

- The OcpsHeader.
- A group identifier (128-bit UUID, generated at group formation).
- An ordered list of member pseudonym-cert IDs.
- The leader cert ID.
- The group mode: FORMING, STABLE, HANDOFF, DISSOLVING.
- The active gap target (seconds) and speed setpoint (m/s).

- A monotonic group-epoch counter (incremented on any membership or leadership change).
- The signature.

8.6 Attestation

Attestation messages convey a signed claim about the integrity of the transmitter's O-CPS software stack and its upstream perception pipeline. Transmission is infrequent (0.1–0.5 Hz). The message contains:

- The OcpsHeader.
- A software identifier (vendor OID, stack version, build fingerprint).
- A hash of the currently-loaded O-CPS configuration.
- A boot-time attestation quote from the HSM or TPM (in the format defined by the platform's attestation service).
- A sensor-stack health summary (count of active sensors by modality, count of degraded sensors).
- The signature.

Receivers use the attestation to apply a trust weight to subsequent PerceptionSummary and Intent messages from the same pseudonym certificate. An attestation older than its lifetime is treated as absent; absence of current attestation is grounds for de-weighting but not for rejection outright, as some conformance classes (notably Class A, §17.1) do not emit attestation.

8.7 Telemetry

Telemetry messages are low-rate, non-safety-critical carriers of data useful to operators and road authorities (anonymized road-condition signals, aggregated congestion estimates). They are explicitly subject to the privacy model of §11 and MUST NOT carry any per-trip identifiers.

9. Trust and Security Model

Security in O-CPS is layered: cryptographic identity provides authentication and non-repudiation; pseudonymity limits long-term correlation; continuous attestation provides integrity signals; misbehavior detection produces consensus-grounded trust scores; and revocation removes compromised or misbehaving participants from the population.

9.1 Identity and Pseudonymous Certificates

9.1.1 Certificate Structure

Every O-CPS participant holds a pool of short-lived pseudonym certificates, each bound to an ECDSA P-256 signing key. Pseudonym certificates **MUST** conform to the profile of IEEE 1609.2.1 (SCMS) or ETSI TS 102 941 (C-ITS). Each certificate carries the Permissions Service Specific Permissions (PSSP) required for the message classes the participant is entitled to transmit; a participant unable to emit a given class **MUST NOT** be issued a certificate that permits it.

9.1.2 Rotation

Certificates **MUST** be rotated at least every 5 minutes during active operation, and a fresh certificate **MUST** be selected on every instance of the following events: entering a new geographic cell (as defined by the SCMS geographic region partition), emerging from a GNSS-denied environment, or power-cycle. Butterfly key expansion per IEEE 1609.2.1 **SHOULD** be used so that pseudonyms within an epoch are not linkable without possession of the CA's linkage keys.

9.1.3 Enrollment

Initial enrollment **MUST** require proof of possession of a hardware root of trust. The vehicle's HSM generates the enrollment keypair; the public key is attested by a manufacturer-issued device certificate; the enrollment certificate authority binds the device attestation to an enrollment identity that is **NEVER** transmitted over the air during normal operation. Only pseudonym certificates travel on the wire.

9.2 Continuous Attestation

Static trust anchors (a valid certificate at session start) do not answer the question of whether a participant's current software and sensor state merits the authority implicit in its messages. O-CPS addresses this with continuous attestation.

A conforming participant emits an Attestation message at least every 10 seconds while actively transmitting PerceptionSummary. The attestation is an HSM- or TPM-signed statement of: (a) the fingerprint of the currently-loaded O-CPS software; (b) the fingerprint of the currently-loaded vehicle perception software; (c) the boot-time attestation of the compute platform; and (d) a summary of the health of each sensor modality.

Peers maintain a per-sender attestation record with a lifetime of 30 seconds. Absence of a current attestation downgrades the sender's trust weight to **ADVISORY** (see §12). A currently-valid attestation whose fingerprints match a known-revoked software build also downgrades the sender, and triggers a misbehavior report.

9.3 Misbehavior Detection

Misbehavior in O-CPS is defined as any pattern of transmitted messages that is internally inconsistent, inconsistent with the laws of physics, or inconsistent with the consensus of peers and local sensors. Each of these is detectable:

- **Internal inconsistency:** a sender reports its own pose at (x,y,z), then reports perceived objects at relative positions that, when transformed to world coordinates, place them on top of its own pose.
- **Physical implausibility:** a sender reports a velocity or acceleration in excess of vehicle-class limits, or reports poses that imply instantaneous teleportation.
- **Consensus deviation:** a sender repeatedly reports objects in positions where no other sender in the same area reports any object, and the receiver's own sensors do not corroborate.
- **Attestation anomaly:** a sender produces valid pseudonym certificates but cannot produce current attestation, or attestation fingerprints correspond to a revoked software build.

A conforming implementation **MUST** run all four classes of detector continuously on all received messages and **MUST** emit a Misbehavior Report over the cellular control-plane interface to the appropriate Misbehavior Authority when any detector fires above its threshold. Misbehavior Reports are OUT of the safety-critical loop; they do not affect real-time fusion. Real-time response is limited to trust-weight de-rating (§12.3).

9.4 Revocation

The SCMS or C-ITS PKI operates a Certificate Revocation List (CRL) and a linkage-based revocation mechanism per IEEE 1609.2.1. O-CPS participants **MUST** ingest the current CRL at least once per day while connectivity permits, and **MUST** refuse to honor messages signed by revoked certificates. Participants whose own enrollment certificate is revoked **MUST** cease all O-CPS transmission within 60 seconds of ingesting the revocation.

10. Privacy Model

Cooperative perception is inherently broadcast. Every PerceptionSummary that leaves a vehicle is received, in principle, by every conformant receiver within radio range. The privacy model is therefore not about confidentiality of individual messages; it is about what those messages are allowed to contain, how they are correlated across time, and what retention any receiver is permitted.

10.1 Data Minimization on the Wire

No O-CPS message, under any circumstance, **MUST** contain:

- The vehicle's Vehicle Identification Number (VIN) or any derived identifier.
- The owner's or driver's name, license number, address, phone number, or any other personally identifiable information.
- A persistent vehicle identifier that survives certificate rotation.
- License plate data, whether the transmitter's own or that of observed vehicles.
- Biometric data of any kind.
- Raw camera imagery or lidar point clouds.

The only identifier on the wire is the pseudonym certificate ID, which rotates per §9.1.2.

10.2 Localization of Data

O-CPS messages are geographically scoped by the GeoNetworking destination area (ITS-G5) or by the PC5 sidelink neighborhood. There is no wide-area broadcast of O-CPS messages and no central subscription service for O-CPS content. Any party that wishes to aggregate O-CPS data across a metropolitan scale **MUST** do so through the Telemetry channel (§8.7), subject to the retention limits below.

10.3 Retention Limits

A conforming O-CPS receiver **MUST** observe the following retention limits:

Data Type	Maximum Retention at Receiver	Exception
PerceptionSummary (others')	60 seconds	None.
Intent (others')	30 seconds	None.
GroupState (foreign groups)	60 seconds	None.
Attestation records	30 seconds after last message from the same cert	None.
Misbehavior evidence	30 days, at Misbehavior Authority only	Receiver MUST purge local copy after report transmission.
Aggregated road condition (Telemetry)	90 days at operator, pre-anonymized before aggregation	Data MUST NOT be joined to identity outside the 30-day enrollment window.

10.4 Opt-In for Telemetry

A vehicle MUST NOT transmit Telemetry messages without explicit owner opt-in. The default configuration from the factory MUST have Telemetry transmission disabled.

PerceptionSummary, Intent, GroupState, and Attestation are NOT subject to opt-in; they are safety-functional and are required for participation in cooperative perception at all.

10.5 Prohibited Analyses

Implementers and operators MUST NOT:

- Attempt to link pseudonym certificate IDs to enrollment identities except through the SCMS linkage mechanism, and only in the course of a properly-authorized misbehavior investigation.
- Construct long-term movement traces of individual vehicles from O-CPS data.
- Sell, trade, or disclose O-CPS-derived data to advertising networks or data brokers.
- Retain Telemetry data beyond the limits in §10.3 without explicit, revocable consent.

11. Fusion and World Model

Each O-CPS participant maintains a World Model: its current estimate of the set of objects in its relevance horizon, fused from its own sensors and from received PerceptionSummary messages. Fusion is the process by which incoming tracks are merged with existing tracks or instantiated as new tracks.

11.1 Fusion Inputs

The fusion engine consumes:

- Local sensor tracks, in the implementation-specific format of the vehicle's perception stack.
- Received PerceptionSummary messages, after signature verification and freshness check.
- Received Intent messages from the same senders, used to prime the motion model for near-term predictions.
- Current trust weights for each sender (§12.3).

11.2 Fusion Rules

A conforming fusion engine MUST observe the following rules:

1. Local sensor tracks at HIGH or VERY_HIGH confidence always dominate over cooperative tracks for the same physical object. A peer message MUST NOT override a confident local detection.
2. A cooperative track with no local corroboration MAY be added to the World Model, but MUST be flagged as cooperative-only and MUST NOT be used as sole input to automated longitudinal or lateral control.
3. When multiple senders report the same physical object, the fused track's state is a trust-weighted average of the reported states; the fused confidence is not higher than the maximum reported confidence, unless the sources are independent (different vendors, different sensor modalities), in which case the fused confidence MAY be one step higher.
4. Stale tracks (no update within the relevant message-class lifetime plus a 50 ms grace window) MUST be aged out.
5. A track whose reported position or velocity is physically implausible given its prior track state MUST be rejected and MUST trigger a misbehavior detector (§9.3).

11.3 Trust Weights

The contribution of a sender to the fused track is scaled by a trust weight in the range [0, 1]. The weight is computed as:

$$w = w_{\text{cert}} * w_{\text{attest}} * w_{\text{history}} * w_{\text{conf}}$$

where

<code>w_cert</code>	= 1.0 if certificate is valid and not on CRL, else 0
<code>w_attest</code>	= 1.0 if attestation is current and fingerprints known-good
	= 0.5 if attestation is absent but cert is valid (Class A)
	= 0.2 if attestation is stale
	= 0.0 if attestation fingerprints are on the revocation list

```
w_history = a rolling average in [0, 1] of the sender's agreement  
            with consensus and with local sensors over the past 5 min  
            (clamped >= 0.1 to allow recovery from transient faults)  
w_conf    = the sender's self-reported confidence, normalized to [0, 1]
```

Any weight of zero in any factor produces a zero total weight and excludes the sender from fusion for that message. Implementations MAY add further factors; MUST NOT remove any of the above.

11.4 Advisory Threshold

A fused track MAY influence driver-visible advisories or ADAS inputs only if its effective trust weight (maximum across contributing sources) is at least 0.5 AND at least one contribution is from a local sensor OR from an attested cooperative source at HIGH confidence or better. Tracks below this threshold MAY be displayed to the driver as informational but MUST NOT gate control decisions.

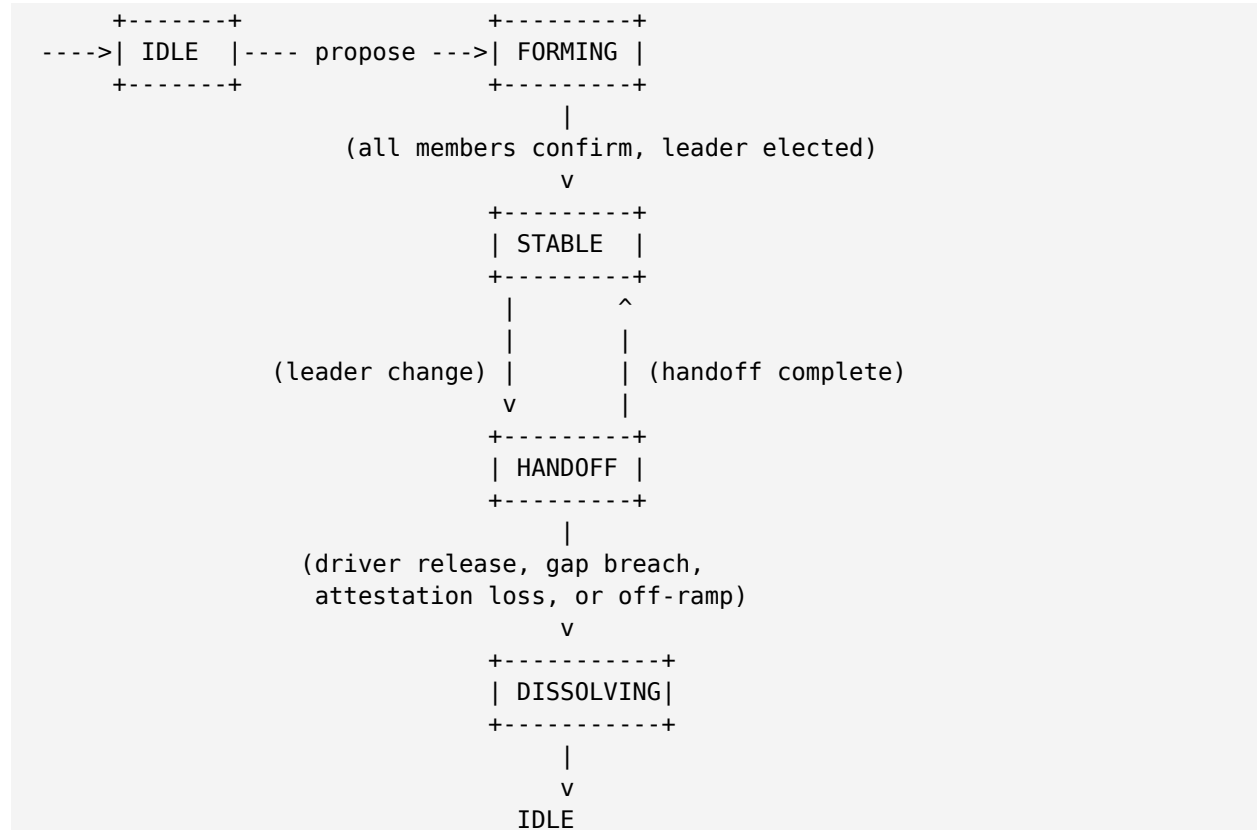
12. Cooperative Behaviors

12.1 C-ACC Group Protocol

Cooperative Adaptive Cruise Control (C-ACC) extends conventional ACC with inter-vehicle coordination. The C-ACC Group Protocol specifies how a set of vehicles forms an explicit coordinated group, maintains synchronized gap targets, hands off leadership, and dissolves.

12.1.1 State Machine

Every C-ACC Group passes through the following states:



12.1.2 Formation

A C-ACC Group is formed by proposal. Any vehicle in an IDLE state MAY transmit a PLATOON_FORM Intent addressed to a specific peer (by pseudonym cert ID) or to the neighborhood. Peers that receive the proposal and wish to join transmit their own PLATOON_FORM Intent within a 2-second window. Once all proposed members have confirmed, one of them generates a group UUID and transmits the first GroupState with mode=FORMING; the group transitions to STABLE once all members have acknowledged the GroupState and the leader has observed stable gap-keeping for at least 5 seconds.

12.1.3 Driver Authorization

No vehicle shall transition from IDLE to FORMING on an inbound proposal without the driver's explicit authorization for that specific proposal. The authorization MUST be revocable at any instant through a standard mechanism on the vehicle's HMI; revocation immediately transitions

the group to DISSOLVING from that vehicle's perspective, and that vehicle's C-ACC function reverts to conventional ACC with the production gap setting.

12.1.4 Leader Election and Handoff

The initial leader of a group is the proposer. Leadership MAY transfer on the following triggers: the leader's intent to leave the group, detection of leader misbehavior by consensus of the remaining members, or an explicit handoff request from the leader. Handoff uses a two-phase commit: the outgoing leader nominates a successor in a GroupState with mode=HANDOFF; the successor accepts with its own GroupState carrying an incremented epoch and the new leader cert ID; once all non-leader members have acknowledged, the group returns to STABLE.

12.1.5 Dissolution

A group dissolves when any of the following occur: all but one member have left; the leader loses attestation or its certificate is revoked; a gap target is violated for more than 3 seconds by any member; or an explicit dissolve is transmitted by the leader. On entry to DISSOLVING, all members revert immediately to local ADAS without cooperative inputs. The group epoch is retired and the group UUID MUST NOT be reused.

12.1.6 Safety Envelopes

Within STABLE, the group's gap target MUST be no shorter than 0.8 seconds on limited-access roads, no shorter than 1.2 seconds on arterial roads, and MUST widen automatically as a function of road class, weather (via local sensors), speed, and the weakest attestation state among group members. A group whose weakest member has lost attestation MUST widen to the conventional ACC production gap within 2 seconds and MUST begin a HANDOFF or DISSOLVING transition as appropriate.

12.2 Slipstream Mode (Optional Extension)

Slipstream Mode is an optional application-layer overlay on top of the C-ACC Group Protocol. It is the Strynex-branded extension referenced in §1.2 and is not required for base O-CPS conformance. This section specifies it because public interoperability requires a common definition.

12.2.1 Functional Additions

- Greenwave integration: consumes SPaT (SAE J2735) messages from intersection RSUs and optimizes the group's approach speed to arrive in the green phase, subject to driver override and posted limits.
- Convoy voice channel: a group-scoped push-to-talk or VOX voice channel using the local sidelink mesh as the primary transport, with cellular fallback. Voice traffic is out-of-band with respect to the safety message classes and MUST NOT contend with PerceptionSummary or Intent for radio resources.
- Role-based cooperative responsibilities: lead-car responsibilities (setting group speed, sharing enhanced perception horizon) and follower responsibilities (maintaining gap, confirming intents).

12.2.2 Transport Preference Order

Slipstream Mode transports, in priority order, are:

6. C-V2X PC5 sidelink or ITS-G5 direct (as for base O-CPS safety messages).
7. Wi-Fi Aware / 802.11bd direct mesh (for voice and non-safety traffic only).

8. Cellular relay via the group leader's uplink (fallback for voice only; MUST NOT be used for any safety message class).

12.3 Intersection Assist (Optional Extension)

Intersection Assist consumes RSU-sourced MAP and SPaT messages (SAE J2735) together with local PerceptionSummary traffic to alert the driver to red-light runners, cross-traffic at blind intersections, and unprotected left-turn conflicts. It produces advisory output only and is always driver-mediated. Detailed specification is out of scope for v0.2 and is deferred to a forthcoming extension profile.

13. Vehicle Integration Interfaces

An O-CPS implementation lives inside a modern vehicle architecture that is already dense with standards. This section specifies how O-CPS integrates without displacing the vehicle's existing safety and security mechanisms.

13.1 Placement in the Vehicle E/E Architecture

O-CPS SHOULD be implemented in a dedicated V2X module or in a high-compute zone controller with physical isolation from the safety-critical powertrain and chassis domains. The V2X module MUST NOT have direct write access to any domain carrying ASIL D signals. All advisory outputs from O-CPS MUST pass through the vehicle's existing ADAS domain controller, which retains authority for actuation decisions.

13.2 In-Vehicle Communication

The V2X module exchanges data with other domains through the following standardized interfaces:

Interface	Purpose	Standard
SOME/IP Service Discovery	Advertise and consume service endpoints across domains over automotive Ethernet.	ISO/SAE 22900 series; AUTOSAR
SecOC (Secure Onboard Communication)	Authenticate safety-relevant signals (e.g., V-CTRL advisories) between the V2X module and ADAS domain.	AUTOSAR SecOC
DoIP	Diagnostic access to the V2X module over Ethernet.	ISO 13400
CAN FD	Legacy interfaces to non-Ethernet domains.	ISO 11898-1:2024
1000BASE-T1	Physical layer for automotive Ethernet where available.	IEEE 802.3bp

13.3 HSM Requirements

The V2X module MUST have access to a Hardware Security Module (HSM) that supports ECDSA P-256 signature generation and verification, secure key storage with anti-extraction, and the attestation operations required for §9.2. The HSM MUST be certified to at least FIPS 140-3 Level 2 or Common Criteria EAL 4+. Private keys MUST never leave the HSM in plaintext form.

13.4 Time Synchronization

Accurate time is foundational. The V2X module MUST maintain a local clock synchronized to UTC to within 10 ms, primarily via GNSS, with a secondary network-time source for GNSS-denied operation. Outbound timestamps use this clock. Inbound timestamps are checked

against it: a message whose timestamp is more than 500 ms in the future or more than its stated lifetime in the past **MUST** be rejected.

14. Safety Architecture

O-CPS is advisory to the vehicle's ADAS, but its advisories are load-bearing in cooperative behaviors. ISO 26262 allocation reflects this: message-layer functions are ASIL B to C depending on class (Table 8-2), while the ADAS retains ASIL D authority for any actual actuation.

14.1 Hazard and Operability Analysis

The following hazards are explicitly in scope of the O-CPS safety analysis. Each is paired with its mitigation.

Hazard	Mitigation
Transmission of fabricated perception	Signed messages, attestation, misbehavior detection, trust-weighted fusion (§9.3, §12.3).
Consumption of spoofed perception	Signature verification, consensus checks, advisory threshold (§12.4), local-sensor dominance (§12.2 rule 1).
Denial of safety message reception	Fail-closed reversion to local ADAS (§4.5); safety-relevant behaviors remain driver-mediated.
Group-state desync within a C-ACC Group	Group epoch counter (§8.5); gap widening on attestation loss (§13.1.6); bounded DISSOLVING transition.
Leader misbehavior within a C-ACC Group	Consensus-based leader-misbehavior detection (§13.1.4); two-phase handoff; dissolution on failure.
Certificate compromise	Short-lived pseudonyms (§10.1.2); CRL ingestion; revocation-triggered transmission halt (§10.4).
GNSS denial or spoofing	Network time fallback (§14.4); local-sensor dominance; attestation-aware gap widening.
Radio congestion under dense deployment	Decongestion per ETSI TS 103 175 (§7.1); priority assignment per class (§9.2); relevance-horizon scoping.

14.2 Fail-Safe State

The canonical fail-safe state for any O-CPS participant is: cease transmission, cease consumption of peer messages for control inputs, revert to purely local-sensor operation with the vehicle's conventional ACC/ADAS settings. Entry into fail-safe MUST be driver-visible, MUST be latchable until operator acknowledgement for safety-critical triggers (misbehavior, certificate revocation), and MUST NOT require cellular connectivity to enter.

14.3 Verification

A conforming implementation SHALL be verified against the ISO 26262 Part 4 and Part 6 lifecycle, and SHALL additionally pass the O-CPS conformance test suite (§17). Verification artifacts SHOULD be published in the vendor's O-CPS conformance report.

15. Deployment Scenarios

Five scenarios exercise the full capability range of the protocol and serve as reference deployments for conformance validation.

15.1 Urban Signalized Intersection

An RSU mounted on a traffic signal transmits SPaT and MAP messages per SAE J2735 and acts as an O-CPS Class B participant: it transmits PerceptionSummary messages derived from roadside sensors (radar, camera) covering the intersection. Approaching vehicles fuse RSU perception with their own, anticipate cross-traffic behind occlusions, and trigger Intersection Assist advisories. Greenwave coordination (§13.2) is optional.

15.2 Limited-Access Highway Platoon

Three to six light passenger vehicles or commercial trucks form a C-ACC Group at highway speeds. The leader's PerceptionSummary extends the effective forward sensor horizon of every follower by 100–400 meters. Fuel-efficiency gains for commercial platoons are a documented secondary benefit but are not claimed by this specification; the normative benefit is safety and anticipation.

15.3 Construction Zone

Temporary RSUs (portable or attached to trucks) broadcast construction-zone MAP deltas and high-confidence PerceptionSummary of workers on foot. Approaching vehicles widen gap targets automatically, and C-ACC Groups dissolve on zone entry if the group's aggregate gap target would be shorter than the zone-posted minimum.

15.4 Vulnerable Road User (VRU) Proximity

A phone or dedicated beacon carried by a pedestrian or cyclist transmits a minimal PerceptionSummary containing its own pose and velocity, at low rate (1–2 Hz) and LOW-to-MEDIUM confidence. Vehicles fuse these into the world model as flagged vulnerable tracks, weighted appropriately. The pseudonymity and retention constraints of §11 apply unconditionally: no vehicle may retain or correlate VRU beacon data beyond 60 seconds.

15.5 Adversarial Conditions

A conformant deployment MUST remain safe under the presence of non-conforming or adversarial participants transmitting valid-signature messages with fabricated content. The mechanisms of §9.3 and §12.3 collectively reduce the impact of an adversary to, at worst, momentary spurious advisories before trust weights decay and misbehavior reports are filed. No adversary holding only a single compromised enrollment SHALL be able to induce unsafe actuation.

16. Reference Implementation and Conformance

16.1 Conformance Classes

O-CPS defines three conformance classes. A compliant implementation **MUST** declare its class in its Attestation messages (or, for Class A, in its PerceptionSummary vendor-extension field).

Class	Capability	Use Case
Class A (Baseline)	Transmits and receives PerceptionSummary and Intent. Does not transmit Attestation. Cannot participate in C-ACC Groups as leader.	Aftermarket retrofit; low-cost OEM integrations; VRU beacons.
Class B (Full)	All of Class A, plus Attestation, GroupState, and C-ACC Group participation (leader or follower). Required for Slipstream Mode.	OEM integrations on vehicles with HSM and automotive Ethernet.
Class I (Infrastructure)	All of Class B, plus support for MAP, SPaT, and environment messages. Extended relevance horizons (up to 500 m).	Roadside units maintained by road authorities and operators.

16.2 Reference Implementation

A reference implementation is maintained under the Strynex program. It consists of a protobuf schema bundle, a C reference encoder/decoder, a Python test harness, and a set of recorded PCAPs covering the five deployment scenarios of §15. The reference implementation is licensed under Apache 2.0 and is intended for conformance testing, not for production deployment.

16.3 Conformance Test Suite

A conforming implementation **MUST** pass the following test categories:

- Wire-format conformance: encoded messages **MUST** decode bit-identically on the reference decoder.
- Cryptographic conformance: signatures **MUST** validate against the reference verifier using SCMS or C-ITS test PKI credentials.
- Timing conformance: rates, latencies, and lifetimes **MUST** be observed within tolerances specified in §8.2.
- Behavioral conformance: state transitions for C-ACC Group formation, handoff, and dissolution **MUST** match the state machine of §13.1.1.
- Safety conformance: fail-safe transitions **MUST** occur under the trigger conditions of §14.2 and complete within 2 seconds.
- Privacy conformance: emitted messages **MUST NOT** contain any of the prohibited fields of §11.1, as verified by automated scan.

- Misbehavior conformance: all four detectors of §9.3 **MUST** be demonstrated against the reference adversarial PCAPs.

17. Standards Alignment and Gap Analysis

O-CPS is deliberately composed from existing standards. This section enumerates what is reused, what is extended, and what O-CPS adds that the underlying standards do not provide.

17.1 Reused

Standard	Reused For
3GPP TS 36.213 / 38.213	C-V2X PC5 PHY/MAC (LTE-V2X and NR-V2X).
IEEE 802.11p / bd	ITS-G5 PHY/MAC.
ETSI EN 302 636	GeoNetworking and BTP over ITS-G5.
SAE J2735 (2024)	BSM, SPaT, MAP, EVA, RTCM message structure.
SAE J3161	PC5 conventions for safety messages.
IEEE 1609.2-2022 / IEEE 1609.2.1	Security profile, SCMS, pseudonym certificates.
ETSI TS 103 097 v2.1.1	Security profile (EU deployment).
ETSI TS 102 940 / TS 102 941	C-ITS PKI architecture and management.
ETSI TS 103 175	Congestion control over ITS-G5.
ISO 26262 (all parts)	Functional safety lifecycle.
AUTOSAR SecOC	In-vehicle signal authentication.

17.2 Extended

- ETSI TS 103 324 CPM is extended with the Normalized Perception Profile (§7), including the normalized object taxonomy and uniform confidence semantics that the CPM base profile leaves to implementation discretion.
- IEEE 1609.2 attestation is extended with a continuous-attestation cadence, the Attestation message class (§8.6), and the trust-weight formula (§12.3).

17.3 Added

- The C-ACC Group Protocol (§13.1). No existing standard specifies a complete state machine for driver-authorized, leaderless-dissolution cooperative cruise control groups.
- The consensus-grounded misbehavior detection framework (§9.3). The underlying IEEE 1609.2 family defines the report format but leaves detector design to implementers; O-CPS specifies four normative detector classes.
- The privacy model (§11). Existing V2X standards permit pseudonymity but do not specify retention limits, prohibit specific data categories, or require opt-in for Telemetry.

18. Governance, Intellectual Property, and Licensing

18.1 Specification License

This specification is published under the Creative Commons Attribution 4.0 International license. Any party MAY reproduce, excerpt, translate, or build derivative specifications from this document, provided they attribute the source and do not assert exclusivity over the underlying specification text.

18.2 Reference Implementation License

The reference implementation (schemas, reference codec, test harness, adversarial PCAPs) is published under the Apache License, Version 2.0. The Apache 2.0 patent grant extends to any contribution accepted into the reference implementation repository.

18.3 Patent Posture

The author, on behalf of Specter Point Intelligence, LLC, asserts no patent claims against good-faith implementations of this specification. Implementers of O-CPS are on notice that third parties may hold patents applicable to the underlying technologies (C-V2X, ITS-G5, SCMS); this specification grants no rights with respect to those patents. A defensive patent pool SHOULD be convened as adoption proceeds.

18.4 Governance Process

Evolution of O-CPS is governed by a lightweight change-proposal process. Any party MAY submit a change proposal to the public repository. Proposals are merged by rough consensus of the active implementer community, with an editorial veto reserved to the editor of record to preserve normative coherence. The intent is to transfer governance to an established standards body (IEEE-SA, ISO TC 22 SC 31, or SAE) once three independent conformant implementations are in public deployment.

18.5 Trademark

“Strynex” is a trademark of Specter Point Intelligence, LLC, applied only to the reference implementation and branded overlay (Slipstream Mode). Use of the term “O-CPS” in conformance claims is governed only by the conformance-class framework of §17 and is not subject to trademark gating.

19. Roadmap

The following milestones define the path from the v0.2 draft to a stable v1.0 release.

Milestone	Deliverables	Target
v0.3 (Interim)	Complete Appendix A schemas in machine-readable protobuf; reference Python test harness; adversarial PCAP corpus.	Q3 2026
v0.4 (Interop)	First public interoperability event with at least two independent implementations; Intersection Assist full specification.	Q1 2027
v0.5 (Infrastructure)	Class I (infrastructure) profile finalized; RSU reference deployment with two participating road authorities.	Q3 2027
v0.6 (Safety)	ISO 26262 safety case template published; independent safety audit of the reference implementation.	Q1 2028
v1.0 (Stable)	Frozen wire format; transfer of governance to an established standards body; public conformance registry.	Q4 2028

20. Conclusion

O-CPS is not an attempt to displace the existing cooperative-perception standards ecosystem. It is a composition of that ecosystem into a single opinionated profile that is implementable today on current radio hardware, auditable against a public reference, and publishable under a license posture that forecloses rent-seeking around the base protocol.

What distinguishes this specification from the existing state of the art is not novelty in any single layer. It is the combination of: a normalized perception profile that makes heterogeneous sensor suites interoperable at the message layer; a continuous-attestation model that makes trust a real-time signal rather than a certificate-issuance event; a driver-authority-absolute posture in every cooperative behavior; and an explicit refusal to introduce cloud dependencies into safety-critical paths.

The standards ecosystem around cooperative perception is crowded. What it has been missing is a composition layer with opinions. This document is that layer. It is offered to implementers, road authorities, and standards bodies as a shared target rather than as a product to be purchased.

Appendix A. Complete Message Schemas

The following protobuf definitions are normative. The package name for all schemas is ocps.v2.

A.1 Common

```
syntax = "proto3";
package ocps.v2;

message OcpsHeader {
  uint32 version      = 1;  // MUST be 2
  uint64 timestamp_ms = 2;
  bytes  cert_id      = 3;  // 8 bytes
  uint32 msg_class    = 4;
  uint32 seq          = 5;
  uint32 horizon_m    = 6;
  uint32 lifetime_ms  = 7;
}

message Pose {
  double lat          = 1;  // WGS-84, degrees
  double lon          = 2;  // WGS-84, degrees
  float  alt_m        = 3;
  float  yaw_deg      = 4;  // clockwise from true north
  float  pitch_deg    = 5;
  float  roll_deg     = 6;
  float  speed_mps    = 7;
  float  accel_mps2   = 8;  // longitudinal, signed
}

message Dimensions {
  uint32 length_cm = 1;
  uint32 width_cm  = 2;
  uint32 height_cm = 3;
}
```

A.2 PerceptionSummary

```
message PerceivedObject {
  uint32 class_id      = 1;  // see §7.2
  uint32 confidence    = 2;  // 0-255, see §7.3
  float  range_m       = 3;
  float  bearing_deg   = 4;  // relative to transmitter heading
  Dimensions dims      = 5;
  float  rel_v_long_mps = 6;
  float  rel_v_lat_mps  = 7;
  uint32 sensor_mask   = 8;  // bit0=radar, bit1=lidar,
                             // bit2=camera, bit3=ultrasonic,
                             // bit4=other
  bytes  vendor_oid    = 9;  // optional; empty unless class_id=0xFF
}

message PerceptionSummary {
```

```

OcpsHeader header    = 1;
Pose      own_pose   = 2;
Dimensions own_dims   = 3;
repeated PerceivedObject objects = 4;
bytes signature      = 15; // ECDSA-P256 over fields 1-4
}

```

A.3 Intent

```

enum IntentCode {
    INTENT_UNKNOWN    = 0;
    STRAIGHT          = 1;
    LANE_CHANGE_LEFT  = 2;
    LANE_CHANGE_RIGHT = 3;
    TURN_LEFT         = 4;
    TURN_RIGHT        = 5;
    MERGE             = 6;
    EXIT              = 7;
    STOP              = 8;
    YIELD             = 9;
    PLATOON_FORM       = 10;
    PLATOON_LEAVE      = 11;
}

message Intent {
    OcpsHeader header    = 1;
    IntentCode code       = 2;
    uint32    prediction_ms = 3;
    uint32    confidence   = 4; // 0-255
    bool      driver_committed = 5;
    bytes     group_id      = 6; // optional, 16 bytes
    bytes     signature     = 15;
}

```

A.4 GroupState

```

enum GroupMode {
    GROUP_MODE_UNKNOWN = 0;
    FORMING             = 1;
    STABLE              = 2;
    HANDOFF            = 3;
    DISSOLVING         = 4;
}

message GroupState {
    OcpsHeader header    = 1;
    bytes     group_id    = 2; // 16 bytes
    repeated bytes members = 3; // ordered; each 8 bytes (cert_id)
    bytes     leader      = 4; // 8 bytes
    GroupMode mode        = 5;
    float     gap_target_s = 6;
    float     speed_setpoint_mps = 7;
    uint64    epoch       = 8;
}

```

```

    bytes    signature    = 15;
}

```

A.5 Attestation

```

message SensorHealth {
    uint32 radar_active      = 1;
    uint32 radar_degraded    = 2;
    uint32 lidar_active      = 3;
    uint32 lidar_degraded    = 4;
    uint32 camera_active     = 5;
    uint32 camera_degraded   = 6;
    uint32 ultrasonic_active = 7;
    uint32 ultrasonic_degraded = 8;
}

message Attestation {
    OcpsHeader header      = 1;
    string vendor_oid      = 2;
    string stack_version    = 3;
    bytes build_fingerprint = 4; // SHA-256
    bytes config_fingerprint = 5; // SHA-256
    bytes platform_quote    = 6; // HSM/TPM attestation blob
    SensorHealth sensors     = 7;
    bytes signature         = 15;
}

```

A.6 Telemetry

```

enum TelemetryKind {
    TELEMETRY_UNKNOWN      = 0;
    ROAD_SURFACE_FRICTION  = 1;
    WEATHER_ESTIMATE       = 2;
    LOCAL_CONGESTION       = 3;
    HAZARD_FLAG            = 4;
}

message Telemetry {
    OcpsHeader header = 1;
    TelemetryKind kind = 2;
    bytes payload = 3; // kind-specific encoding
    bytes signature = 15;
}

```

Appendix B. Glossary of Acronyms

Acronym	Expansion
ADAS	Advanced Driver Assistance System
ASIL	Automotive Safety Integrity Level (ISO 26262)
BSM	Basic Safety Message (SAE J2735)
BTP	Basic Transport Protocol (ETSI)
C-ACC	Cooperative Adaptive Cruise Control
C-ITS	Cooperative Intelligent Transportation System (EU)
C-V2X	Cellular Vehicle-to-Everything (3GPP)
CAM	Cooperative Awareness Message (ETSI)
CPM	Collective Perception Message (ETSI TS 103 324)
CRL	Certificate Revocation List
DoIP	Diagnostics over Internet Protocol (ISO 13400)
ECDSA	Elliptic Curve Digital Signature Algorithm
ECU	Electronic Control Unit
EVA	Emergency Vehicle Alert (SAE J2735)
GNSS	Global Navigation Satellite System
HARQ	Hybrid Automatic Repeat Request
HMI	Human-Machine Interface
HSM	Hardware Security Module
ITS-G5	European 5.9 GHz DSRC radio stack (IEEE 802.11p/bd)
MAP	Map Data Message (SAE J2735)
MBD	Misbehavior Detection
NR-V2X	New Radio V2X (3GPP Rel-16+)
O-CPS	Open Cooperative Perception System (this specification)
OID	Object Identifier
PC5	Direct device-to-device sidelink interface (3GPP)
PPPP	ProSe Per-Packet Priority
PSM	Personal Safety Message (SAE J2735)
PSSP	Permissions Service Specific Permissions

Acronym	Expansion
RSU	Roadside Unit
SCMS	Security Credential Management System (US)
SecOC	Secure Onboard Communication (AUTOSAR)
SOME/IP	Scalable service-Oriented MiddlewarE over IP
SPaT	Signal Phase and Timing (SAE J2735)
TPM	Trusted Platform Module
UUID	Universally Unique Identifier
V2X	Vehicle-to-Everything
VRU	Vulnerable Road User
WGS-84	World Geodetic System 1984

Appendix C. Normative References

- 3GPP TS 36.213: Evolved Universal Terrestrial Radio Access (E-UTRA); Physical layer procedures.
- 3GPP TS 38.213: NR; Physical layer procedures for control.
- 3GPP TS 23.285: Architecture enhancements for V2X services.
- IEEE 802.11-2020: Wireless LAN Medium Access Control and Physical Layer Specifications.
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